

CPSC 304 Project Cover Page

Milestone #: 4

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Group Number: 83

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

AI declaration:

No AI tools were used in the process of completing Milestone 4.

1. A short description of the final project, and what it accomplished.

This project is a model for the management and moderation of a massively multiplayer online role-playing game (MMORPG). The database represents core game entities such as players, NPCs, items, quests, clans, locations, and more. The project allows moderators to perform basic moderation, such as adding/updating enemy NPCs, removing items, and looking up moderation related information, with comprehensive player search functionality. Additional functionality includes player and NPC look-ups at a location, and detailed clan breakdowns.

2. A description of how your final schema differed from the schema you turned in. If the final schema differed, explain why

No changes to schema were made from the previous milestones.

3. A visual representation of the state of your database after running the .sql file.

Profession:

ProfessionID	Name	Wage	MinStats
1	Blacksmith	120	HP: 0, ATK: 12, DEF: 8, SPD: 0
2	Alchemist	110	HP: 0, ATK: 0, DEF: 4, SPD: 6
3	Hunter	100	HP: 0, ATK: 10, DEF: 0, SPD: 10
4	Merchant	90	HP: 0, ATK: 0, DEF: 2, SPD: 8
5	Miner	95	HP: 10, ATK: 8, DEF: 10, SPD: 0

Class:

ClassID	Name	MinLevel	MaxMana
1	Warrior	1	100
2	Mage	1	220
3	Archer	1	140
4	Priest	1	200
5	Rogue	1	120

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Ability:

ClassID	Name	ManaCost	Cooldown	BaseDamage	BufsToBaseStats
1	Shield Bash	20	8	35	HP: 0, ATK: 6, DEF: 4, SPD: -1
2	Fireball	35	5	60	HP: 0, ATK: 10, DEF: 0, SPD: 0
3	Piercing Arrow	25	4.5	45	HP: 0, ATK: 8, DEF: -2, SPD: 2
4	Holy Light	30	6	25	HP: 25, ATK: 0, DEF: 3, SPD: 0
5	Backstab	18	3.5	50	HP: 0, ATK: 12, DEF: 0, SPD: 4
1	Fast Blade	30	12	20	HP: 0, ATK: 10, DEF: 0, SPD: 0
1	Whirl wind	40	10	55	HP: 0, ATK: 8, DEF: -2, SPD: 0
2	Super Nova	45	8	50	HP: 0, ATK: 0, DEF: 0, SPD: -4
2	Butane Blast	50	6	75	HP: 0, ATK: 12, DEF: 0, SPD: 0
3	Rain of Arrows	35	9	60	HP: 0, ATK: 6, DEF: 0, SPD: 0
3	Eagle Eye	20	7	40	HP: 0, ATK: 10, DEF: -1, SPD: 3
4	Divine Shield	40	15	0	HP: 50, ATK: 0, DEF: 10, SPD: 0
4	Smite	25	4	40	HP: 0, ATK: 8, DEF: 0, SPD: 0
5	Shadow Step	22	5	35	HP: 0, ATK: 0, DEF: 0, SPD: 10
5	Poison Blade	28	6.5	45	HP: 0, ATK: 8, DEF: 0, SPD: 2

Clan:

ClanName	MinLevelToJoin	ClanRank	NumMembers
DragonSlayers	10	1	25
MoonGuard	5	2	18
ShadowLeaf	8	3	14
IronLegion	12	4	20
Wanderers	1	5	30
Super Awesome Cool Guy Squad	1	6	6

GlobalEvent:

EventID	Name	TimeStart	TimeEnd	EXP_Reward	CurrencyReward
1	Harvest Festival	2026-03-01 8:00:00	2026-03-07 23:59:59	300	150
2	Goblin Invasion	2026-03-10 10:00:00	2026-03-12 22:00:00	500	250
3	Lunar Eclipse	2026-03-15 20:00:00	2026-03-16 2:00:00	450	200
4	Arena Week	2026-03-18 9:00:00	2026-03-25 23:59:59	350	300
5	Winter Hunt	2026-03-28 6:00:00	2026-04-03 23:59:59	400	220

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Quest:

QuestID	Name	MinLevel	EXP_Reward	CurrencyReward	EventID
1	Defend the Village	5	120	60	2
2	Collect Herbs	3	80	40	3
3	Win 3 Arena Matches	10	200	120	4
4	Hunt Wolves	8	160	90	5
5	Deliver Supplies	1	50	25	1
6	Find Jerry	1	20	30	NULL

Location:

LocationID	Name	Terrain	LocalTime	LocationLevel	Weather
1	Greenhaven	Plains	Morning	1	Sunny
2	Ashen Peak	Mountain	Noon	12	Windy
3	Silverwood	Forest	Evening	7	Rainy
4	Frostfang Keep	Tundra	Night	15	Snow
5	Sunken Marsh	Swamp	Dusk	10	Foggy

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NPC:

NPC_ID	Name	NPC_Level	BaseStats	LocationID
1	Gruk	6	HP: 120, ATK: 18, DEF: 8, SPD: 6	1
2	Emberfang	14	HP: 260, ATK: 40, DEF: 20, SPD: 12	2
3	Nightclaw	9	HP: 180, ATK: 24, DEF: 12, SPD: 16	3
4	Frostmaw	16	HP: 300, ATK: 45, DEF: 25, SPD: 8	4
5	Boglurker	11	HP: 210, ATK: 28, DEF: 14, SPD: 10	5
6	Saul Goodman	60	HP: 1500, ATK: 300, DEF: 67, SPD: 167	1
7	Elder Dragon	55	HP: 10000, ATK: 99, DEF: 420, SPD: 35	2
8	Atakhan	35	HP: 4400, ATK: 60, DEF: 130, SPD: 48	5
9	Gromp	8	HP: 240, ATK: 9, DEF: 10, SPD: 4	5
10	Treant	18	HP: 500, ATK: 29, DEF: 34, SPD: 10	3
11	Hobgoblin	9	HP: 200, ATK: 22, DEF: 10, SPD: 8	1
12	Orc	13	HP: 333, ATK: 40, DEF: 25, SPD: 6	1
13	Jerry	3	HP: 50, ATK: 6, DEF: 3, SPD: 9	1
14	Wind Spirit	10	HP: 80, ATK: 17, DEF: 8, SPD: 30	2
15	Bandit	15	HP: 250, ATK: 31, DEF: 17, SPD: 24	3
16	Bandit Leader	20	HP: 350, ATK: 44, DEF: 40, SPD: 24	3
17	Fenrir	46	HP: 2500, ATK: 85, DEF: 100, SPD: 150	4
18	Snowfox	14	HP: 222, ATK: 30, DEF: 13, SPD: 28	4
19	Winter Rabbit	12	HP: 167, ATK: 19, DEF: 11, SPD: 25	4
20	Lost Spirit	15	HP: 300, ATK: 32, DEF: 25, SPD: 20	5
21	Gustavo Fring	27	HP: 666, ATK: 66, DEF: 66, SPD: 66	5
22	Troll	19	HP: 444, ATK: 46, DEF: 30, SPD: 27	5
23	Ogre	23	HP: 750, ATK: 99, DEF: 67, SPD: 13	5

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EnemySpecies:

EnemySpecies	SpawnRate	Aggression	EnemyType
Goblin	0.8	0.65	Humanoid
Fire Drake	0.25	0.9	Dragon
Dire Wolf	0.55	0.75	Beast
Ice Troll	0.3	0.85	Giant
Swamp Slime	0.7	0.4	Elemental
Elder Dragon	0	0.99	Dragon
Atakhan	0.01	0.99	Elemental
Giant Frog	0.95	0.25	Beast
Treant	0.15	0.66	Plant
Spirit	0.45	0.6	Elemental
Bandit	0.5	0.5	Humanoid
Bandit Leader	0	0.6	Humanoid
Fenrir	0	0.3	Beast
Fox	0.4	0.21	Beast
Rabbit	0.63	0.12	Beast
Troll	0.33	0.7	Humanoid
Ogre	0.2	0.65	Humanoid

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Enemy:

NPC_ID	EnemySpecies	EXPDropped	GoldDropped
1	Goblin	25	10
2	Fire Drake	120	55
3	Dire Wolf	45	18
4	Ice Troll	140	65
5	Swamp Slime	60	22
7	Elder Dragon	10000	100000
8	Atakhan	1000	780
9	Giant Frog	40	5
10	Treant	170	37
11	Goblin	50	15
12	Goblin	99	31
14	Spirit	57	14
15	Bandit	111	256
16	Bandit Leader	350	2148
17	Fenrir	2500	8000
18	Fox	121	49
19	Rabbit	101	41
20	Spirit	133	88
22	Troll	207	123
23	Ogre	369	131

Item:

ItemID	Name	ItemType	BaseCost
1	Iron Sword	Weapon	150
2	Oak Staff	Weapon	140
3	Hunter Bow	Weapon	145
4	Blessed Dagger	Weapon	135
5	War Hammer	Weapon	170
6	Preserved Heirloom	Artifact	500
7	Health Potion	Consumable	50
8	Mana Potion	Consumable	100
9	Ornate Jewelry	Artifact	1200
10	Thick Club	Weapon	90

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Weapon:

ItemID	WeaponStats	MinStats	BuffsToBaseStats
1	HP: 0, ATK: 20, DEF: 0, SPD: 0	HP: 0, ATK: 10, DEF: 0, SPD: 0	HP: 0, ATK: 2, DEF: 0, SPD: 0
2	HP: 0, ATK: 16, DEF: 0, SPD: 2	HP: 0, ATK: 6, DEF: 0, SPD: 4	HP: 0, ATK: 3, DEF: 0, SPD: 1
3	HP: 0, ATK: 18, DEF: 0, SPD: 4	HP: 0, ATK: 8, DEF: 0, SPD: 8	HP: 0, ATK: 2, DEF: 0, SPD: 2
4	HP: 0, ATK: 17, DEF: 0, SPD: 6	HP: 0, ATK: 7, DEF: 0, SPD: 9	HP: 0, ATK: 2, DEF: 0, SPD: 3
5	HP: 20, ATK: 24, DEF: 2, SPD: -2	HP: 0, ATK: 14, DEF: 4, SPD: 0	HP: 10, ATK: 3, DEF: 1, SPD: -1
10	HP: 0, ATK: 28, DEF: -5, SPD: -5	HP: 30, ATK: 20, DEF: 0, SPD: 0	HP: 15, ATK: 10, DEF: 0, SPD: 0

Drops:

NPC_ID	ItemID	ItemDropRate
1	4	0.2
20	6	0.18
22	7	0.1
11	8	0.17
5	1	0.18
2	2	0.15
23	10	0.12
16	8	0.4
12	7	0.5
3	3	0.25
15	9	0.05
4	5	0.1
7	9	1
1	7	0.15
15	8	0.1
16	6	0.6
12	10	0.1
11	7	0.08
15	6	0.15
23	7	0.35
16	9	1

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PlayerLevel:

PlayerLevel	ClassID	BaseStats
5	1	HP: 180, ATK: 18, DEF: 14, SPD: 6
8	2	HP: 110, ATK: 8, DEF: 6, SPD: 10
7	3	HP: 130, ATK: 16, DEF: 8, SPD: 18
10	4	HP: 140, ATK: 10, DEF: 12, SPD: 10
12	5	HP: 150, ATK: 20, DEF: 10, SPD: 22
3	1	HP: 100, ATK: 9, DEF: 7, SPD: 4
6	4	HP: 80, ATK: 5, DEF: 5, SPD: 6
5	3	HP: 90, ATK: 10, DEF: 5, SPD: 13
9	3	HP: 185, ATK: 26, DEF: 12, SPD: 23
27	5	HP: 500, ATK: 83, DEF: 35, SPD: 73
15	4	HP: 320, ATK: 17, DEF: 21, SPD: 16
19	4	HP: 600, ATK: 25, DEF: 39, SPD: 21
11	5	HP: 135, ATK: 18, DEF: 9, SPD: 20
14	2	HP: 230, ATK: 47, DEF: 13, SPD: 28
14	3	HP: 420, ATK: 61, DEF: 26, SPD: 32
45	1	HP: 1500, ATK: 191, DEF: 223, SPD: 56

Player:

Username	PlayerLevel	Currency	Mana	LastSeenOnline	ProfessionID	ClassID	ClanName	LocationID
todd	8	500	160	2026-03-05 18:20:00	2	2	MoonGuard	3
jonsnow	5	220	55	2026-03-06 9:10:00	1	1	Wanderers	1
aegon	7	340	85	2026-03-04 21:45:00	3	3	ShadowLeaf	3
walterwhite	10	620	150	2026-03-06 11:00:00	4	4	DragonSlayers	2
jesse	12	710	95	2026-03-05 23:30:00	5	5	IronLegion	5
awesomeguy1	3	18	100	2026-03-22 21:40:00	3	1	Super Awesome Cool Guy Squad	1
awesomeguy2	6	312	21	2026-03-22 21:30:00	2	4	Super Awesome Cool Guy Squad	1
awesomeguy3	5	343	17	2026-03-22 21:30:00	4	3	Super Awesome Cool Guy Squad	1
awesomeguy4	9	565	175	2026-03-22 21:45:00	1	3	Super Awesome Cool Guy Squad	5
Michael Scott	6	677	67	2026-03-25 16:00:00	5	4	Super Awesome Cool Guy Squad	5
TheLegend27	27	329	600	2026-03-30 12:00:00	1	5	NULL	5
Tyler Ninja Blevins	15	2581	240	2026-03-19 19:45:00	1	4	DragonSlayers	3
johngamer	19	1188	321	2026-04-01 21:44:30	4	4	NULL	3
aaaaaa	11	246	180	2026-03-31 2:30:00	4	5	MoonGuard	2
kaicenat	14	923	46	2026-03-17 21:01:00	4	2	ShadowLeaf	5
ishowspeed	14	895	103	2026-03-17 21:00:00	5	3	ShadowLeaf	5
awesomeguy5	10	1234	123	2026-03-23 22:00:00	4	4	Super Awesome Cool Guy Squad	2
fortnite	45	24020	1200	2026-04-02 22:00:00	2	1	DragonSlayers	4

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InInventory:

Username	ItemID
todd	2
jonsnow	1
aegon	3
walterwhite	5
jesse	4
aaaaaa	6
jonsnow	8
aegon	7
Tyler Ninja Blevins	10
awesomeguy4	7
awesomeguy3	7
awesomeguy1	1
kaicenat	5
awesomeguy2	2
fortnite	9
ishowspeed	4
Michael Scott	8
Michael Scott	7
walterwhite	4
fortnite	10

IsActive:

Username	QuestID
todd	2
jonsnow	5
aegon	1
walterwhite	3
jesse	4

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IsFriendsWith:

Player1	Player2	
todd	aegon	
todd	walterwhite	
jonsnow	aegon	
aegon	jesse	
walterwhite	jesse	
Michael Scott	awesomeguy1	
Michael Scott	awesomeguy2	
Michael Scott	awesomeguy3	
Michael Scott	awesomeguy4	
Michael Scott	awesomeguy5	
awesomeguy1	Michael Scott	
awesomeguy1	awesomeguy3	
awesomeguy1	awesomeguy4	
awesomeguy1	awesomeguy2	
awesomeguy2	awesomeguy3	
awesomeguy3	awesomeguy4	
awesomeguy1	awesomeguy5	
awesomeguy3	awesomeguy5	
walterwhite	Tyler Ninja Blevins	
fortnite	walterwhite	
ishowspeed	kaicenat	
ishowspeed	aegon	
Michael Scott	aaaaaa	
fortnite	johngamer	
johngamer	todd	
Tyler Ninja Blevins	kaicenat	
Tyler Ninja Blevins	ishowspeed	
fortnite	aaaaaa	

4. A list of all SQL queries used to satisfy the rubric items and where each query can be found in the code (file name and line number(s))

Non-Hardcoded Queries:

- Insert query:

CPSC304_Node_Project-main/appService.js
insertEnemy
Lines 294 - 324

Variables: enemySpecies, expDropped, goldDropped, npcId

```
INSERT INTO Enemy (NPC_ID, EnemySpecies, EXPDropped, GoldDropped)
VALUES (npcId, enemySpecies, expDropped, goldDropped)
```

-Update query:

CPSC304_Node_Project-main/appService.js
updateEnemy
Lines 220 - 248

Variables: enemySpecies, expDropped, goldDropped, npcId

```
UPDATE Enemy
SET EnemySpecies = enemySpecies,
    EXPDropped = expDropped,
    GoldDropped = goldDropped
WHERE NPC_ID = npcId
```

-Delete query:

CPSC304_Node_Project-main/appService.js
deleteItem
Lines 327 - 339

Variables: itemId

```
DELETE FROM Item WHERE ItemID = itemId
```

-Selection query:

CPSC304_Node_Project-main/appService.js
selectPlayers
Lines 366 - 433

```
SELECT p.Username, p.PlayerLevel, pl.BaseStats, p.Currency, p.Mana,
       p.LastSeenOnline, p.ProfessionID, prof.Name, p.ClassID,
       c.Name, p.ClanName, loc.LocationID, loc.Name
FROM Player p, PlayerLevel pl, Location loc, Profession prof, Class c
```

```
WHERE (p.ProfessionID = prof.ProfessionID) AND  
      (p.ClassID = c.ClassID) AND  
      (p.LocationID = loc.LocationID) AND  
      (p.PlayerLevel = pl.PlayerLevel) AND  
      (p.ClassID = pl.ClassID)
```

-Projection query:

```
CPSC304_Node_Project-main/appService.js  
questProjection  
Lines 342 - 362
```

```
SELECT ordering FROM Quest
```

Where ordering is any amount of: QuestID, Name, MinLevel, EXP_Reward,
CurrencyReward, EventID

-Join query:

```
CPSC304_Node_Project-main/appService.js  
getPlayerAbilities  
Lines 250 - 266
```

Variables: username

```
SELECT p.Username, c.Name AS ClassName, a.Name AS AbilityName,  
       a.ManaCost, a.Cooldown, a.BaseDamage  
FROM Player p  
JOIN Class c ON p.ClassID = c.ClassID  
JOIN Ability a ON c.ClassID = a.ClassID  
WHERE p.Username = username
```

Hardcoded Queries:

5. For SQL queries 7 through 10 inclusive, include a copy of your SQL query and a maximum of 1-2 sentences describing what that query does.

- *Aggregation with group by:*

File Name: CPSC304_Node_Project-main/appService.js

Line Number: 530-534

```
`SELECT loc.LocationID, loc.Name, COUNT(*)  
      FROM Location loc, Player p  
      WHERE loc.LocationID = p.LocationID  
      GROUP BY loc.LocationID, loc.Name  
      ORDER BY COUNT(*) DESC`
```

This query groups all players by their current location and returns the location ID, location name, and total player count at each location. This query allows moderators to identify which locations have a lot of players and which locations have few players.

- *Aggregation with having:*

File Name: CPSC304_Node_Project-main/appService.js
Line Number: 499-516

```
SELECT c.ClanName, c.MinLevelToJoin, c.ClanRank,  
       COUNT(p.Username) AS NumMembers,  
       MIN(p.PlayerLevel) AS MinPlayerLevel,  
       MAX(p.PlayerLevel) AS MaxPlayerLevel,  
       ROUND(AVG(p.PlayerLevel), 2) AS AvgPlayerLevel  
FROM Clan c  
JOIN Player p ON c.ClanName = p.ClanName  
GROUP BY  
  c.ClanName,  
  c.MinLevelToJoin,  
  c.ClanRank,  
  c.NumMembers  
HAVING COUNT(p.Username) >= :minPlayerCount
```

This query groups players by clan and calculates each clans member count, min, max and average player level and then returns only clans whose actual player count is equal or greater than the specified minimum count.

- *Nested aggregation with group by:*

File Name: CPSC304_Node_Project-main/appService.js
Line Number: 548-555

```
SELECT npc.NPC_ID, npc.Name, npc.NPC_Level,  
       npc.BaseStats, loc.LocationID, loc.Name  
FROM Location loc, NPC npc  
WHERE loc.LocationID = npc.LocationID AND npc.NPC_Level > ALL  
      (SELECT AVG(npc2.NPC_Level)
```

```
FROM NPC npc2
GROUP BY npc2.LocationID)
ORDER BY npc.NPC_Level DESC
```

This query finds NPCs that are higher level than the average NPC level at every location. The subquery finds the average npc level grouped by location, and the main query selects the NPCs.

- *Division*

File Name: CPSC304_Node_Project-main/appService.js

Line Number: 271-284

```
SELECT p.Username, p.ClanName, cl.ClanRank, cl.NumMembers
FROM Player p
JOIN Clan cl ON p.ClanName = cl.ClanName
WHERE p.ClanName IS NOT NULL
AND (SELECT COUNT(*) FROM Player p3 WHERE p3.ClanName =
p.ClanName) > 1
AND NOT EXISTS (
    SELECT p2.Username
    FROM Player p2
    WHERE p2.ClanName = p.ClanName
    AND p2.Username != p.Username
    MINUS
    SELECT f.Player2
    FROM IsFriendsWith f
    WHERE f.Player1 = p.Username)
```

This query finds players who are in a clan with more than one member and are friends with all other players in their clan. It does that by checking whether a player is missing any clanmates in their friends list then keeps only the players where none are missing.